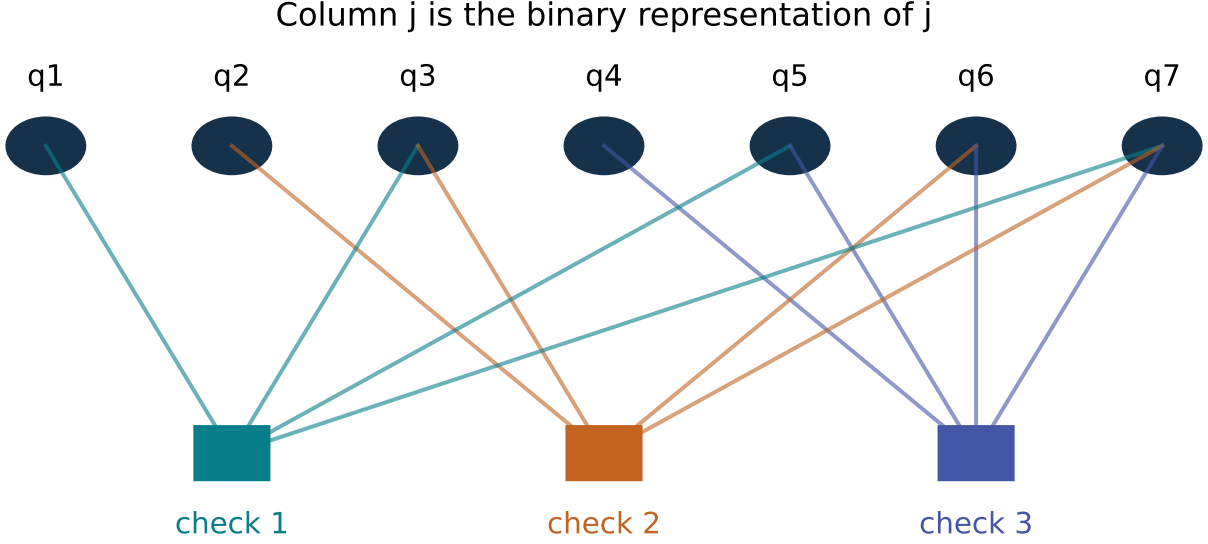
Column j is the binary representation of j
q1
q2
q3
q4
q5
q6
q7
check 1
check 2
check 3